



DFDOF

Forensic Baseline Report

Drone Forensic Decision and Orchestration Framework

Case Identifier:	ft-df049-2018-apr-ios
Operator:	Floris
Analysis Started:	2026-06-18 12:19:28
Analysis Completed:	2026-06-18 12:21:02
Evidence Directory:	C:\Users\Floris\Documents\dji_drones\train_mavic_air\df049\2018_april\ios
Output Directory:	C:\Users\Floris\Documents\dji_drones\train_mavic_air\df049\2018_april\dfdof_output\ft-df049-2018-apr-ios
Evidence Sources:	2
Pipeline Version:	DFDOF v1.0 (proof of concept)

This report is an automated forensic baseline produced by DFDOF. It is not a court-ready expert report. All findings are bounded by the data available and must be interpreted by a qualified forensic examiner. The populated case output directory is the authoritative source and provides the full overview of the pipeline's findings.

1. Evidence Intake and Chain of Custody

The following input evidence files were ingested. SHA-256 and SHA-1 hashes were computed at intake to establish integrity baselines. No modification to original evidence files is performed by DFDOF.

File	Classification	Acquisition	Size	Hash Timestamp
df049_internal.E01	drone_sd	physical	7.8 GB	2026-06-18 12:19:55
ios_logical.zip	controller_ios	logical	23.7 MB	2026-06-18 12:19:55

Evidence Source Path

C:\Users\Floris\Documents\ddj\ddrones\train_mavic_air\df049\2018_april\ios

Cryptographic Hashes

df049_internal.E01		drone_sd	physical
SHA-256:	ce48d2b0d1b634442795a80c804053ef25bbaa6013e97eacc166458331a4a8d8		
SHA-1:	4663934cb481d453bd7afe709c81069fff5adde		

ios_logical.zip		controller_ios	logical
SHA-256:	b1f904faeaae2a190f4e1a3a6e91c8a63c540c530e2d12a82f154c95a4a90667		
SHA-1:	882c874b51138546d4f382459db5ea751af9366b		

Pipeline Execution Summary

Completed Phases

1. p1_provenance
2. p2_image_parsing
3. p3_artefact_extraction
4. p4_decision_and_orchestration
5. p5_normalisation_and_anomaly_checking
6. p6_multisource_correlation
7. p7_analysis_and_validation
8. p8_automated_reporting

Anomaly Flags

1. [p1 - drone sd]: mmls: no partition table found in df049_internal.E01; treating as direct partition image
2. [p2 - controller android]: source evidence not found
3. [p3 - controller android]: source evidence not found
4. [p3 - controller ios - drone logs]: no artefacts found
5. [p3 - controller ios - flight records]: no artefacts found
6. [p3 - controller ios - images]: no artefacts found
7. [p3 - controller ios - videos]: no artefacts found
8. [p3 - drone flight storage]: source evidence not found
9. [p4 - controller android]: source evidence not found
10. [p4 - drone flight storage]: source evidence not found

Tool Execution Status

Tool	Invocations	Successes	Status
exiftool	10	10	[OK]
fls	1	1	[OK]
icat	10	10	[OK]
mmls	1	0	[X] Failed
parser_ios	1	1	[OK]

Processing Coverage Score

Metric	Result
Evidence Sources Detected	2 / 4
Artefact Categories With Data	5 / 7
Flights With Primary Correlation	0 / 0
Tools Succeeded	4 / 5

3. Device and Account Identification

Field	Value	Confidence	Sources
Installed DJI App(s)	com.dji.go	[K] Controller-only	1
Controller Device	iPad mini 4	[K] Controller-only	1
Controller Device	iPad5,1__weibosdk__003143000__iphone__os 10.3.2	[K] Controller-only	1
DJI Application Version	4.2.12	[K] Controller-only	1
Account E-Mail	Droneforensics@vtoinc.com	[K] Controller-only	1
Product Type	38	[K] Controller-only	1
Last Country Code	US	[K] Controller-only	1
Last App Launch	2018-04-19T17:50:21Z	[K] Controller-only	1
Last Flight Date	1970-01-01T00:00:00Z	[K] Controller-only	1

5. Artefact Coverage

Source Coverage

Source	Detected
controller_android	No
controller_ios	Yes
drone_sd	Yes
drone_flight_storage	No

Artefact Counts per Source and Category

Source	Account Data	Databases	Drone Logs	Flight Logs	Flight Records	Images	Videos
controller_ios	1	7	0	2	0	0	0
drone_sd	0	0	0	0	0	6	4

Data Quality Notes per Category

Databases (7 files):

- 3 database(s) empty (controller_ios)

Flight Logs (2 files):

- formats: 2 bracketed_log

Images (6 files):

- 6 file(s) missing (normalisable) EXIF timestamp
- 6 file(s) missing EXIF GPS coordinates

Videos (4 files):

- 4 file(s) missing (normalisable) EXIF timestamp
- 4 file(s) missing EXIF GPS coordinates

6. Data Quality and Anomalies

Data Quality Observations

- Flight_logs: log files present in the following format(s): 2 bracketed_log.

Pipeline Anomaly Flags

1. [p1 - drone sd]: mmls: no partition table found in df049_internal.E01; treating as direct partition image
2. [p2 - controller android]: source evidence not found
3. [p3 - controller android]: source evidence not found
4. [p3 - controller ios - drone logs]: no artefacts found
5. [p3 - controller ios - flight records]: no artefacts found
6. [p3 - controller ios - images]: no artefacts found
7. [p3 - controller ios - videos]: no artefacts found
8. [p3 - drone flight storage]: source evidence not found
9. [p4 - controller android]: source evidence not found
10. [p4 - drone flight storage]: source evidence not found

Artefact-Level Anomaly Detail

The following summarises which automated checks fired for each artefact processed in Phase 5. Row counts indicate affected data rows; column lists are truncated to five entries.

Databases

djiFMDB.db

- database_empty: all tables empty

flysafe_app_dynamic_areas.db

- database_empty: all tables empty

flysafe_license.db

- database_empty: all tables empty

Flight Logs

upgradeLog_0.txt

- format: bracketed_log | entries: 14

upgradeLog_1.txt

- format: bracketed_log | entries: 6

Images

DJI_0001.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0002.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0003.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0004.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0005.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0006.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

- exif_contains_no_gps: is/are missing EXIF GPS coordinates

Videos

DJI_0001.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0004.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0005.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0006.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

7. Further Investigation Items

The following items were identified by automated analysis as requiring manual investigator follow-up. They represent unresolved questions or limitations that the pipeline cannot resolve automatically.

1. Note: no drone serial number found - Device and account information.
2. Note: no drone model found - Device and account information.
3. Note: no firmware version found - Device and account information.
4. Note: no last known latitude (find aircraft) found - Device and account information.
5. Note: no last known longitude (find aircraft) found - Device and account information.
6. Note: no last known altitude (find aircraft) found - Device and account information.
7. Note: no last known heading (find aircraft) found - Device and account information.
8. Note: no last known horiz. accuracy (find aircraft) found - Device and account information.
9. 4 database file(s) contain records and should be examined for relevant artefacts.

Appendix A - Extracted Artefact Hash Manifest

Each artefact extracted in Phase 3 is listed below with its full SHA-256 and SHA-1 cryptographic hashes. These values can be used to independently verify the integrity of every extracted file.

com.dji.go.plist		Account Data	controller_ios	extract_logical
SHA-256:	c6098d9afd1a9df4d1d130eff689a5976e5091b8c0f89be7c59dfecb4ab783b4			
SHA-1:	37a854907f9e26c1732ace247280d91cbe08a13c			

djiFMDB.db		Databases	controller_ios	extract_logical
SHA-256:	c2f956359b0006c3cf0a1ecad392d8c70da65f2ef9145fa6b995b84653329638			
SHA-1:	9d77797e6788fe25f2d71c838c7d59eae0164680			

dji_force_update.db		Databases	controller_ios	extract_logical
SHA-256:	9eb3ad22602803c66f875460d8ccf602f73dc95c0f873fc560cc42bfa1bd8b15			
SHA-1:	8ed825a4d39d14b5b6ff9434a2ac1c3e3090808a			

flysafe_app_dynamic_areas.db		Databases	controller_ios	extract_logical
SHA-256:	7ca264ba8538a4ed920c02222c3898714c0a8199da67bcf029b0e9ed2a36f6ab			
SHA-1:	759c3d9368daf2a307dcc53729c6eccc40fcab66			

flysafe_areas_djigo.db		Databases	controller_ios	extract_logical
SHA-256:	d9164be112208833fe06583920459508675742a54e3ca70d358789ab362c7d3c			
SHA-1:	936e7b4a02e38dc7cd2b2d4cb7be8280096b17bd			

flysafe_dji_flight_dynamic_areas.db		Databases	controller_ios	extract_logical
SHA-256:	8f4020031c32a9185fa05884e52ecaf75c7cdd4e9f9558bae4c2378080943830			
SHA-1:	0cc24ed0288110e2bb9e27d1ca48bf25b8d35416			

flysafe_license.db		Databases	controller_ios	extract_logical
SHA-256:	ef071ebc84923187a5c8819ddf39b58a160aaebd0e8f75a7e41a0fcc90c9da33			
SHA-1:	037c6f40d49371285301094fd840cfde0ee41633			

flysafe_static_circle.db		Databases	controller_ios	extract_logical
SHA-256:	4787fc859c865dab6b3d0db8eb68a55c764ca5d13105b14429cc2d455708cfb4			
SHA-1:	ccf2d5995e798abe1f1255c252c28ec8ae699dd4			

upgradeLog_0.txt		Flight Logs	controller_ios	extract_logical
SHA-256:	2db1db909483d3834493779a1be6fb940ec5d3a68c77570ca8b2c75ace85909d			
SHA-1:	c56c2650c79c1e6f7f5dec7d6c4401a1383c7b0d			

upgradeLog_1.txt		Flight Logs	controller_ios	extract_logical
SHA-256:	e4e7914f4e8a1f49abab9bfeaf450937d93ee4c2514e1c89d591232e25708315			
SHA-1:	1f1e1e999d8af1726611094e55edf3e32769a087			

DJI_0001.THM		Images	drone_sd	extract_physical
SHA-256:	7b7d8a5f04de4720da9c1d65924375c0a19425649b715c9be9d95f431ffe48ef			
SHA-1:	4b066c18aff86e0949ee2429e960119a060b5ad4			

DJI_0002.JPG		Images	drone_sd	extract_physical
SHA-256:	a859aac8fa7aef573e8ae8ae3880d8f3ee02b1325af52634cb8409c1afa94303			
SHA-1:	7b1610a6d6348d5fa470b9d560ca5233305c4cec			

DJI_0003.JPG		Images	drone_sd	extract_physical
SHA-256:	d1dcc50fa8edd7d2c5533bfa132ab0ce83711bcd731b2618ac375493204c98db			
SHA-1:	0eb3c492ab8c8ede7b38c21cf79a61735e6ec7f5			

DJI_0004.THM		Images	drone_sd	extract_physical
SHA-256:	da822f8e97c7453374509b92551f91fc5823f61c969d8f372600bf034ffa3feb			
SHA-1:	0b4527095c4e76844df4dc7c3a6177486c2fa103			

DJI_0005.THM		Images	drone_sd	extract_physical
SHA-256:	ba8631c371d71a3ff5a502566ebdf3772e25a1b88794300f943035795106b0e7			
SHA-1:	d33a50de0b692f83b445759004924cb4cf0e636e			

DJI_0006.THM		Images	drone_sd	extract_physical
SHA-256:	45f1f4e8ca1a8e9a319a7c16ac93ff020646e480121e85cf9768363713777e41			
SHA-1:	8ab6ad224bc3360a44ac2380500a15435abccbb8			

DJI_0001.MP4		Videos	drone_sd	extract_physical
SHA-256:	1457d7112e401e547986c4f0c172f896353d62e4b5d973c949b5353fc1c55778			
SHA-1:	95ca55621f71eb8e63873546809c13e97d753715			

DJI_0004.MP4		Videos	drone_sd	extract_physical
SHA-256:	efab6df387144c83cf98e56c8c83bf15db89d4d810f8b9afc7f57df2e004c095			
SHA-1:	5b51e7202edd35a65ce852c88fc5da0fe2248b31			

DJI_0005.MP4		Videos	drone_sd	extract_physical
SHA-256:	5a82b3a1d0bab3d575fd3a0884bd34012b0744368e672bf9bf568793021c44cf			
SHA-1:	032c213f233635ae0e9407b448d5fc80cbd70e80			

DJI_0006.MP4		Videos	drone_sd	extract_physical
SHA-256:	fdda7dbce2012b8258cde0832897d72373d0882a311936fdd4772c0af14edb83			
SHA-1:	68d706e06c209a9717a152cbc5a9eef5f615eb17			

Appendix B - Evidence Derivation Summary

This table summarises how many artefacts were derived at each pipeline phase, per evidence source and category. It demonstrates the processing chain from extracted artefacts (P3) through decoded outputs (P4) to normalised analysis-ready files (P5).

Controller Ios

Category	P3 Extracted	P4 Decoded	P5 Normalised
Account Data	1	1	0
Databases	7	0	0
Flight Logs	2	0	0

Drone Sd

Category	P3 Extracted	P4 Decoded	P5 Normalised
Images	6	6	4
Videos	4	4	0

Appendix C - Tool Invocation Log

All external tool invocations executed during this analysis are recorded below, in chronological order. Each entry includes the timestamp, tool name, version, return code, full command, and output files produced.

2026-06-18 12:19:28 mmls vThe Sleuth Kit ver 4.15.0 rc=1

cmd: mmls.exe df049_internal.E01

2026-06-18 12:19:28 fls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: fls.exe -r -p df049_internal.E01

2026-06-18 12:20:07 parser_ios v- rc=n/a

cmd: ios_logical.zip C:\Users\Floris\Documents\dfdof_output\ft-df049-2018-apr-ios\p2_image_parsing\controller_ios_parsed
out: controller_ios_parsed

2026-06-18 12:20:08 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df049_internal.E01 1357829
out: DJI_0001.MP4

2026-06-18 12:20:08 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df049_internal.E01 1357830
out: DJI_0002.JPG

2026-06-18 12:20:08 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df049_internal.E01 1357832
out: DJI_0003.JPG

2026-06-18 12:20:20 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df049_internal.E01 1357834
out: DJI_0004.MP4

2026-06-18 12:20:35 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df049_internal.E01 1357835
out: DJI_0005.MP4

2026-06-18 12:20:49 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df049_internal.E01 1357836
out: DJI_0006.MP4

2026-06-18 12:20:58 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df049_internal.E01 1358853
out: DJI_0001.THM

2026-06-18 12:20:58 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df049_internal.E01 1358854
out: DJI_0004.THM

2026-06-18 12:20:59 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df049_internal.E01 1358855
out: DJI_0005.THM

2026-06-18 12:20:59 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df049_internal.E01 1358856
out: DJI_0006.THM

2026-06-18 12:20:59 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0001.THM
out: DJI_0001_exif.json

2026-06-18 12:20:59 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0002.JPG
out: DJI_0002_exif.json

2026-06-18 12:21:00 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0003.JPG
out: DJI_0003_exif.json

2026-06-18 12:21:00 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0004.THM
out: DJI_0004_exif.json

2026-06-18 12:21:00 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0005.THM
out: DJI_0005_exif.json

2026-06-18 12:21:00 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0006.THM
out: DJI_0006_exif.json

2026-06-18 12:21:01 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0001.MP4
out: DJI_0001_exif.json

2026-06-18 12:21:01 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0004.MP4
out: DJI_0004_exif.json

2026-06-18 12:21:02 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0005.MP4
out: DJI_0005_exif.json

2026-06-18 12:21:02 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0006.MP4
out: DJI_0006_exif.json

Appendix D - DFDOF Proof of Concept Overview

The Drone Forensic Decision and Orchestration Framework (DFDOF) is an automated 8-phase forensic pipeline for DJI drone evidence. It ingests images of the controller, drone flight storage and SD cards, then processes them into a structured analysis and report.

P1 Provenance and Integrity

Classifies input files (ZIP/E01/.001) by content signatures. Computes SHA-256 and SHA-1 for all input evidence. Presents a source classification summary for operator confirmation before processing begins.

Input: evidence directory. Output: classified evidence objects with hashes.

P2 Image Parsing

Makes all sources search-ready. Decrypts iOS backups into a domain directory tree. Extracts Android device metadata (DeviceInfo.xml, ApplicationInfo.xml, packages.list). Produces backup_info.json with device identifiers and installed DJI applications.

Input: raw evidence files. Output: parsed directory structures, backup_info.json.

P3 Artefact Extraction

Root-driven category extraction. Discovers DJI app roots by bundle-ID segment matching. Recurses into configured per-platform category paths and filters by extension. Rejects empty files. Each artefact is wrapped in an Evidence object with a parent hash linking it to its source.

Input: parsed directories. Output: categorised artefacts (flight records, drone logs, flight logs, databases, images, videos, account data).

P4 Decision and Orchestration

Dispatches each artefact category to the appropriate specialist tool: DatCon for .DAT drone logs (produces CSV + KML), TXTlogToCSV for flight record .txt files, ExifTool for images and videos, Python parsers for account data (.plist/.xml). Computes hashes for all tool outputs.

Input: extracted artefacts. Output: decoded CSVs, KML files, EXIF JSON, account data observations.

P5 Normalisation and Anomaly Checking

Normalises all CSVs to UTC timestamps and a unified schema. Applies 9 shared telemetry anomaly checks (timestamp regression/gaps, zero coordinates, GPS quality, altitude spikes, motor-airborne-off) plus format-specific checks for flight records (battery voltage, temperature) and drone logs (quaternion, hDOP). Detects empty databases, unreadable flight logs, and missing (normalisable) EXIF timestamps.

Input: decoded artefacts. Output: normalised CSVs, anomaly observations.

P6 Multi-Source Correlation

Correlates flight records with drone logs using two criteria: (1) temporal overlap ≥ 60 s AND $\geq 90\%$ of the shorter flight's duration; (2) median haversine distance of nearest-neighbour GPS pairs ≤ 25 m with ≥ 5 matched pairs. Deduplicates FR candidates when multiple apps recorded the same flight. Builds a unified per-flight event timeline from all matched sources. Also correlates media files by EXIF timestamp, GPS bounding box, and video duration. In the correlation summary: Overlap is the temporal overlap as a percentage of the shorter flight's duration; GPS Match Rate is the fraction of flight-record GPS rows that found a spatially matched drone-log row within a ± 2 s window.

Input: normalised CSVs. Output: timeline_flightXX.json per flight.

P7 Analysis and Validation

Aggregates results from all prior phases into a structured analysis. Resolves device identity (drone serial, model, app version, account email) across sources with confidence levels (controller-only / corroborated / inconsistent). Computes per-flight metrics (peak height, media capture, record completeness). Produces coverage scores and a grouped conclusions structure with a dedicated further_investigation section.

Input: all phase outputs. Output: analysis dict with conclusions in state.json.

P8 Automated Reporting

Generates this PDF forensic baseline report from all prior phase outputs. Produces sensor telemetry plots (altitude, speed, attitude, battery, GPS) from normalised CSVs, a dual-source flight track plot, and the structured appendices. All figures are also saved as standalone PNG files.

Input: state.json (all phase outputs). Output: PDF report + PNG plot files.

Disclaimer

DFDOF was developed by Floris Krijger, MSc Security and Network Engineering, University of Amsterdam, as a Master's thesis project (1 April 2026 - 30 June 2026). This proof of concept has been developed and tested on a targeted subset of the NIST VTO forensic drone dataset and is not a finished, production-ready system. Before use in any operational or legal context, further testing, validation across additional drone models and evidence types, and robustness improvements are required. The authors accept no liability for conclusions drawn from automated pipeline output without independent expert verification.